

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – 9

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Class: TYIT	Batch: 1
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Aim : Cleaning and Structuring Data for Analysis

A) Identify and handle missing or inconsistent data:

Step 1: Load the Dataset

- . Open Tableau Desktop and load the given dataset.
- . Go to Sheet 1.

Step 2: Clean Physics and Attendance

- . Right-click in the Data Pane - Create Calculated Field.

. Create Physics_Clean.

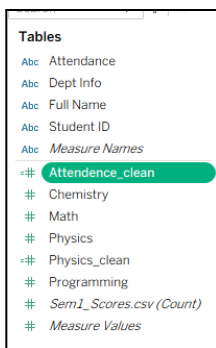
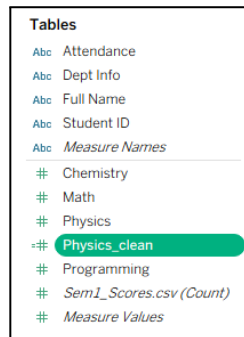
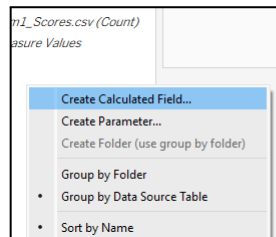
Float (REPLACE (STR ([Physics]), '%', ''))

. Repeat the same steps and create Attendance_Clean.

Float (TRIM (REPLACE (STR ([Attendance]), '%', '')))

Physics_clean

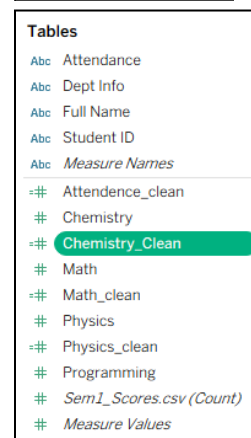
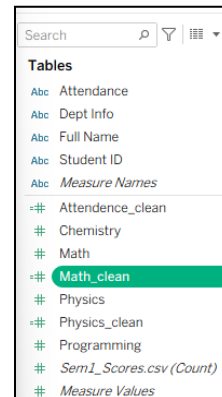
Float (REPLACE (STR ([Physics]), '%', ''))



Step 3: Handle Missing Values

. Create Math_Clean using IFNULL([Math], 0).

. Create Chemistry_Clean using IFNULL([Chemistry], 0).



Chemistry_Clean

IFNULL([Chemistry], 0)

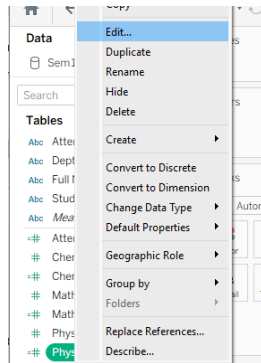
Math_clean

IFNULL([Math], 0)

Step 4: Handle Missing Physics Values

- . Right-click Physics_Clean - Edit and use:
Formula: IFNULL(FLOAT (REPLACE (STR([Physics])), '%', '')), 0)

[Physics]], '%', '')), 0)



Physics_clean

IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)

- . Select Number (decimal).

B) Convert Wide Data into Tall Format:

Step 1: Open Data Source . Open the Data Source tab.

Step 2: Pivot Columns:

- . Hold Ctrl and select Math, Physics, Chemistry, Programming.
- . Right-click the selected columns - Pivot.

#	Math	
#	Physics	Create Calculated Field...
#	Chemistry	Pivot
#	Programming	Merge Mismatched Fields

Field Name	Physical Tab
Dept Info	Sem1_Score
Attendance	Sem1_Score
Pivot Field Names	Pivot
Pivot Field Values	Pivot
Physics_clean	Calculation
Attendance_clean	Calculation
Math_clean	Calculation

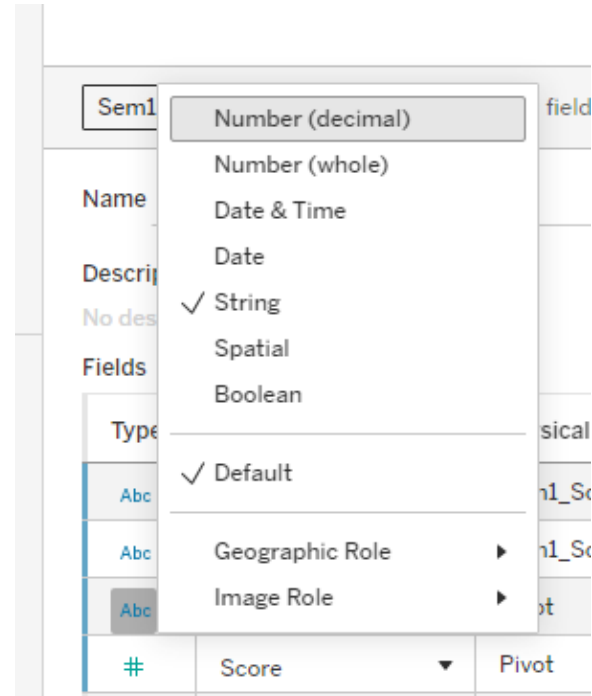
Step 3: Rename Pivoted Fields

- . Rename Pivot Field Names to Subject.
- . Rename Pivot Field Values to Score.

Abc	Attendance	S
Abc	Subject	P
#	Score	P
=#	Physics_clean	C
=#	Attendance_clean	C

Step 4: Format Score

- . Click the Abc icon above Score.

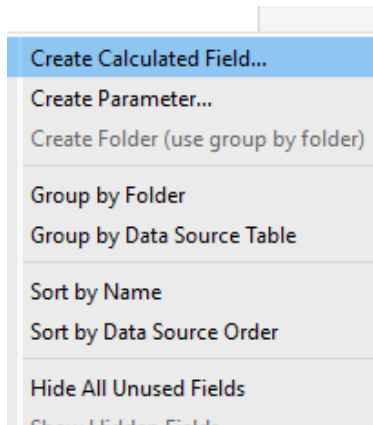


Alternative: Create Clean_Score

Formula: IFNULL (FLOAT (REPLACE (STR([ore]), '%', '')), 0)

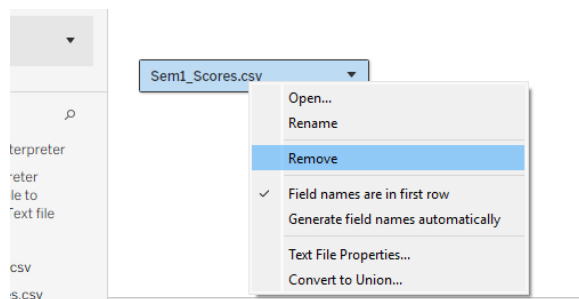
Clean_Score

```
IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)
```



C) Perform Union of Multiple Files:

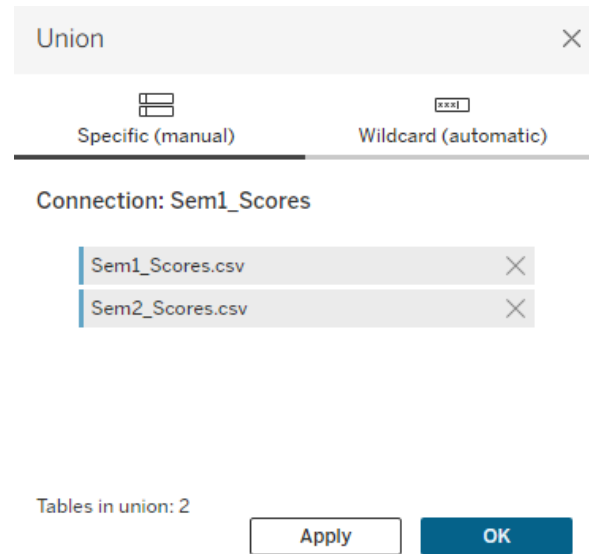
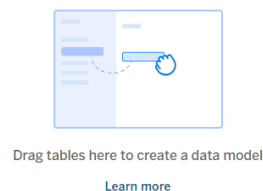
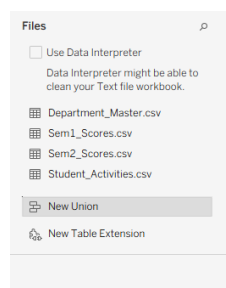
Step 1: Open the Data: Clear the existing canvas



Step 2: Create a Multi-File Union

. Drag New Union to the canvas.

. Add Sem1_Scores.csv and Sem2_Scores.csv.



Step 3: Verify the Merged Data Grid

. Check the data grid at the bottom.

. Confirm that the total is 20 rows.

. Check the Table Name column for the two files.



3. Scroll to the far right of the grid to verify the generated system column: Table Name: Displays 'Sem1_Scores.csv' for the first 10 rows and

"Sem2_Scores.csv" for the remaining 10 rows.

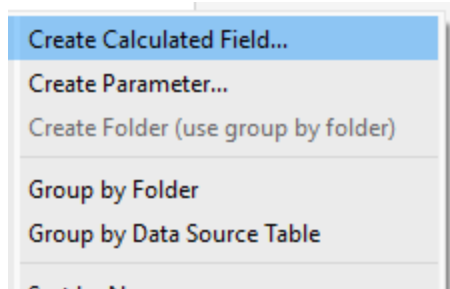
	Abc Union Table Name	Calculation Physics_clean	Calculation Attendance
%	Sem1_Scores.csv	0.900000	
%	Sem1_Scores.csv	0.750000	
%	Sem1_Scores.csv	0.820000	
%	Sem1_Scores.csv	0.880000	
%	Sem1_Scores.csv	0.000000	
%	Sem1_Scores.csv	0.910000	
%	Sem1_Scores.csv	0.700000	

Step 4: Create a Clean Semester Identifier
nension

1. Click on Sheet 1 at the bottom-left.

2. Create Calculated Field.Set the Field Name to:
nester

4. Enter the following formula:



Semester

```
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END
```

D) Execute cross-database joins:

Step 1: Stay on the Top Canvas Leave your Union table box as it is on the canvas.

Step 2: Add and Link Department_Master.csv

1. In the left sidebar under Files / Connections, find Department_Master.csv. 2. Drag Department_Master.csv onto the canvas and drop it to the right of your Union box.

3. A flexible line (relationship / join noodle) will connect the two boxes.



4. If the relationship dialog opens (or click the relationship line):

- Left Table Field (Union): Select Dept_Info

- Right Table Field (Department_Master): Select Dept_Code

Operator: =

Union — Departme...

Name Department_Master.csv

How do relationships differ from joins? [Learn more](#)

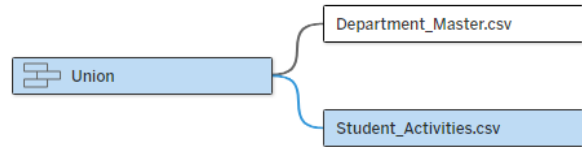
Union	Operator	Department_Maste...
Abc Dept_Info	=	Abc Dept Code



Step 3: Add and Link Student_Activities.csv

1. From the left sidebar, drag Student_Activities.csv onto the canvas.

2. Drop it to the right of the Union box.



3. In the relationship dialog:

- Left Table Field (Union): Select Student_ID

- Right Table Field (Student_Activities): Select Student_ID

Operator: =

Union — Student_A...

Name Student_Activities.csv

How do relationships differ from joins? [Learn more](#)

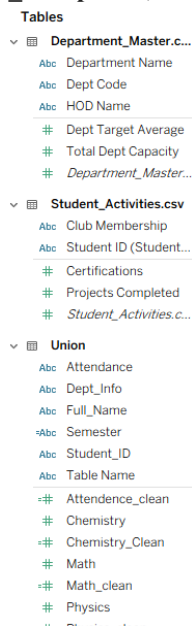
Union	Operator	Student_Activities....
Abc Student_ID	=	Abc Student ID (Stu...

Step 4: Verify in Sheet 1

1. Click on Sheet 1 at the bottom-left.

2. In the Data Pane on the far left, you will now all three tables organized with their respective fields:

- Union: Student_ID, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name
- Department_Master: Dept_Code, Department_Name, HOD_Name, Dept_Target_Average, Total_Dept_Capacity
- Student_Activities: Club_Membership, Projects_Completed, Certifications



E. Restructure Data for Better Visualization

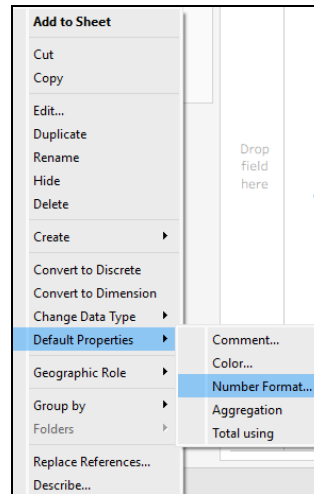
Step 1: Calculate Total Marks

1. Open Tableau.
2. Go to Sheet 1.
3. In the Data Pane, right-click on an empty area.
4. Select Create Calculated Field.
5. Enter field name:
Total_Score
6. Enter formula:
[Math_Clean] + [Physics_Clean] +
[Chemistry_Clean] + [Programming]
7. Click OK.



Step 2: Calculate Overall Percentage

1. Right-click in the Data Pane.
2. Select Create Calculated Field.
3. Field name:
Overall_Percentage
4. Formula:
[Total_Score] / 400.0
5. Click OK.
6. Right-click Overall_Percentage.
7. Select Default Properties → Number Format.
8. Select Percentage.
9. Set decimal places to 1.
10. Click OK.

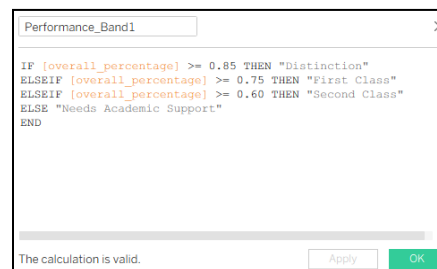


Step 3: Create Performance Band

1. Right-click in Data Pane.
2. Select Create Calculated Field.
3. Field name: Performance_Band

Enter:

```
IF [Overall_Percentage] >= 0.85 THEN "Distinction"  
ELSEIF [Overall_Percentage] >= 0.75 THEN "First  
Class"  
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second  
Class"  
ELSE "Needs Academic Support"  
4. END  
5. Click OK.
```



Step 4: Create Attendance Status

1. Right-click Data Pane.
2. Select Create Calculated Field.
3. Field name:
Attendance_Status

Formula: IF [Attendance_clean] >= 75 THEN
"Eligible (Good Standing)"
ELSE
"Shortage Warning (<75%)"
END

4. END
5. Click OK.



Step 5: Build Student Performance Scorecard

Now create the visualization:

1. Drag Student_ID → Rows.
2. Drag Full_Name → Rows.
3. Drag Semester → Columns.
4. Drag Overall_Percentage → Text in Marks card.
5. Drag Performance_Band → Color.
6. Drag Attendance_Status → Tooltip or Detail.

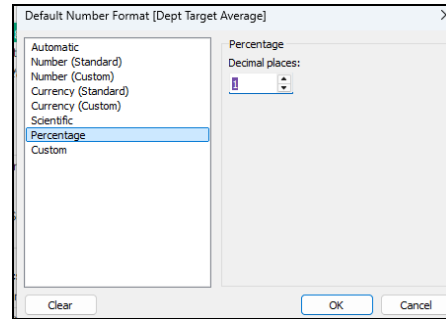
Output:-

Student ID	Full Name	Semester	Overall_Percentage
S101	Aarav Sharma	r.2	67.2%
S102	Priya Verma		
S103	Rohan Gupta		
S104	Sneha Patel		69.7%
S105	Aditya Rao		
S106	Ananya Singh		66.5%
S107	Rahul Nair		
S108	Kavya Reddy		71.5%
S109	Vikram Joshi		49.4%
S110	Pooja Iyer		63.0%

F. Handle Different Levels of Detail Using LOD

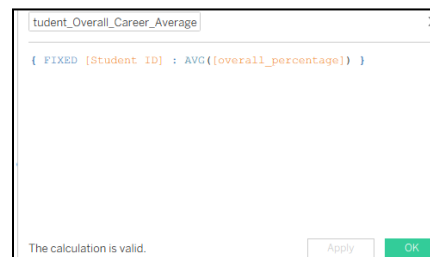
Step 1: Department Average

1. Go to Sheet 1 or create Sheet 2.
2. Right-click in Data Pane.
3. Select Create Calculated Field.
4. Field name: Dept_Average_Percentage_LOD
5. Formula: { FIXED [Dept_Info] : AVG([Overall_Percentage]) }
6. Click OK.
7. Right-click the new field.
8. Select Default Properties → Number Format.
9. Select Percentage and 1 decimal place.



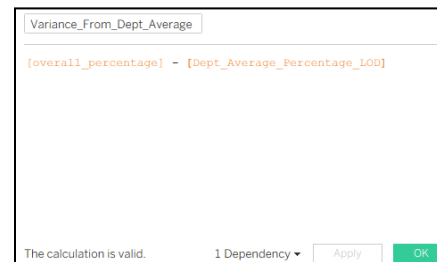
Step 2: Student Career Average

1. Right-click Data Pane.
2. Select Create Calculated Field.
3. Field name: Student_Overall_Career_Average
4. Formula: { FIXED [Student_ID] : AVG([Overall_Percentage]) }
5. Click OK.
6. Set the number format to Percentage with 1 decimal place.



Step 3: Calculate Variance

1. Create another Calculated Field.
2. Field name: Variance_From_Dept_Average
3. Formula: [Overall_Percentage] - [Dept_Average_Percentage_LOD]
4. Click OK.
5. Set it to Percentage with 1 decimal place.



Step 4: Create Benchmark Status

1. Create another Calculated Field.
2. Field name: Benchmark_Status

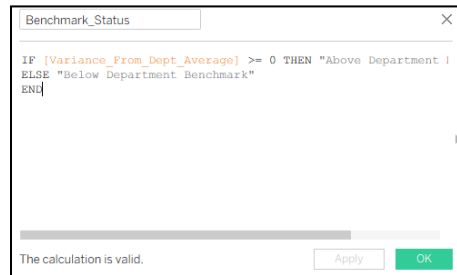
Formula:

IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"

ELSE "Below Department Benchmark"

3. END

4. Click OK.



Step 5: Build LOD Visualization

1. Drag Dept_Info → Rows.
2. Drag Student_ID → Rows.
3. Drag Full_Name → Rows.
4. Drag Semester → Columns.
5. Drag Overall_Percentage → Text.
6. Drag Dept_Average_Percentage_LOD → Tooltip.
7. Drag Benchmark_Status → Color.

Final result

Pages

Columns

Semester

Rows

Dept Info

Student ID

Full Name

Filters

Automatic

Color

Size

Text

Dual

Tooltip

Benchmark_S1

SUM(Dept_T

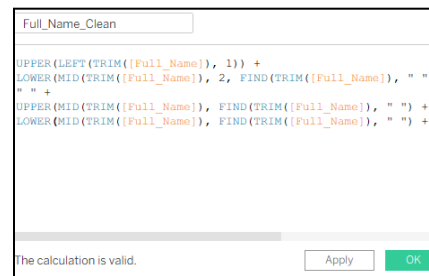
SUM(Overall_p

Build LOD Visualization

Dept Info	Student ID	Full Name	Semester	r 2
CSE-01	S101	Aarav Sharma	67.2%	
	S103	Rohan Gupta		
	S106	Ananya Singh	66.5%	
	S109	Vinay Joshi	49.4%	
ECE-02	S102	pritya VERMA	63.0%	
	S107	Rahul Nair		
	S110	Pooja Iyer	63.0%	
IT-03	S104	Sneha Patel	69.7%	
	S108	Kavya Reddy	71.5%	
MECH-04	S105	aditya Rao		

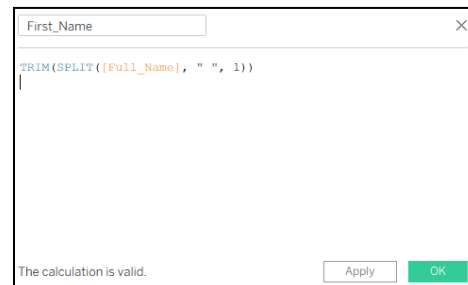
LOWER(MID(TRIM([Full_Name]),
FIND(TRIM([Full_Name]), " ") + 2,
LEN(TRIM([Full_Name]))))

6. OK click karo.



STEP 2 — First_Name

1. Data Pane → Right Click
2. Create Calculated Field
3. Name: First_Name
4. Formula: TRIM(SPLIT([Full_Name], " ", 1))
5. OK.



STEP 3 — Last_Name

1. Data Pane → Right Click
2. Create Calculated Field
3. Name: Last_Name
4. Formula: TRIM(SPLIT([Full_Name], " ", -1))
5. OK.

G: Apply Advanced Data Cleaning Techniques

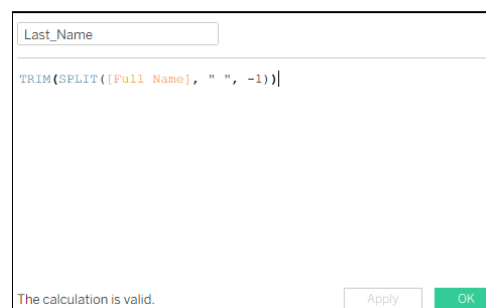
STEP 1 — Full_Name_Clean

1. Sheet 1 par raho.
2. Left side Data Pane mein right-click karo.
3. Create Calculated Field select karo.
4. Field Name: Full_Name_Clean
5. Formula:

UPPER(LEFT(TRIM([Full_Name]), 1)) +
LOWER(MID(TRIM([Full_Name]), 2,
FIND(TRIM([Full_Name]), " ") - 1)) +
" " +

UPPER(MID(TRIM([Full_Name]),
FIND(TRIM([Full_Name]), " ") + 1, 1)) +

STEP 4
—



Dept_Branch

Tumhare Dept_Info mein values jaise CSE-01, ECE-02 hain.

1. Create Calculated Field

2. Name:

Dept_Branch

SPLIT([Dept_Info], "-", 1)

4. OK.

A screenshot of a dialog box titled 'Dept_Branch'. The title bar has a close button (X). The main area contains the formula `SPLIT([Dept_Info], "-", 1)`. At the bottom, there is a status bar that says 'The calculation is valid.' followed by '1 Dependency' and two buttons: 'Apply' and 'OK'.

STEP 5 — Section_ID

1. Create Calculated Field

2. Name:

Section_ID

3. Formula:

SPLIT([Dept_Info], "-", 2)

4. OK.

A screenshot of a dialog box titled 'Section_ID'. The title bar has a close button (X). The main area contains the formula `SPLIT([Dept_Info], "-", 2)`. At the bottom, there is a status bar that says 'The calculation is valid.' followed by two buttons: 'Apply' and 'OK'.

STEP 6 — Student_Display_Label

1. Create Calculated Field

2. Name:

Student_Display_Label

3. Formula:

[Student_ID] + " - " + [Full_Name_Clean] + " (" +
[Dept_Branch] + ")"

4. OK.

A screenshot of a dialog box titled 'Student_Display_Label'. The title bar has a close button (X). The main area contains the formula `[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"`. At the bottom, there is a status bar that says 'The calculation is valid.' followed by two buttons: 'Apply' and 'OK'.